

1.

$$\frac{dx}{dt} = ax + 3x^2$$

equilibrium points

$$ax + 3x^2 = 0$$

$$(a + 3x)x = 0$$

$$x = 0$$

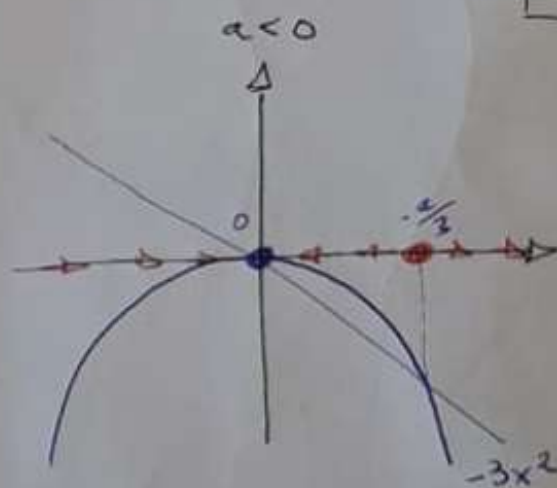
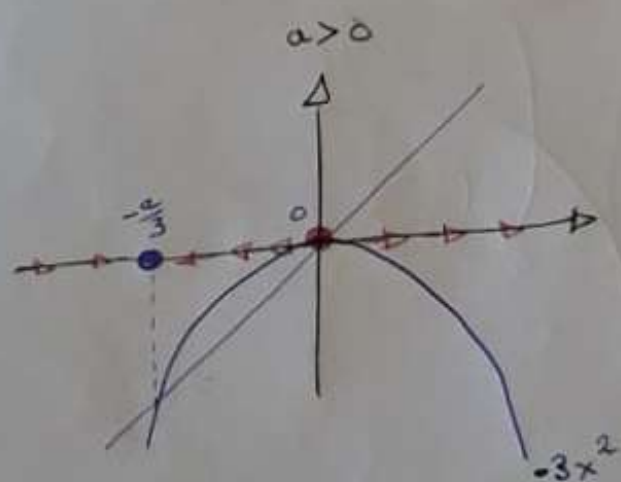
$$x = -\frac{a}{3}$$

stability

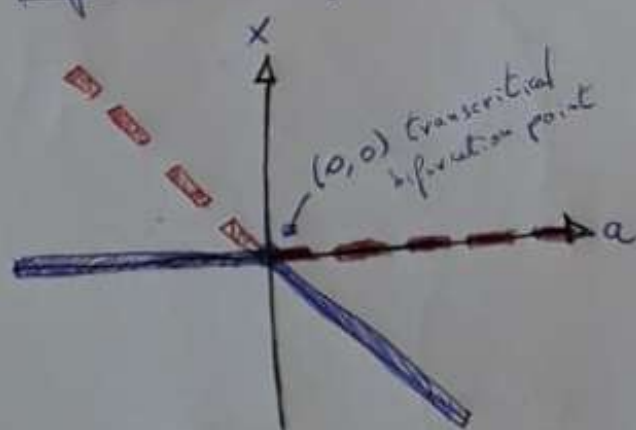
$ax + 3x^2 > 0 \Rightarrow$ trajectory goes right ($ax > -3x^2$)

$ax + 3x^2 < 0 \Rightarrow$ trajectory goes left ($ax < -3x^2$)

• : unstable
• : stable



bifurcation diagram



--- : unstable
— : stable

2. $\frac{dx}{dt} = ax + 2x^3$

equilibrium points

$$ax + 2x^3 = 0$$

$$(a + 2x^2)x = 0$$

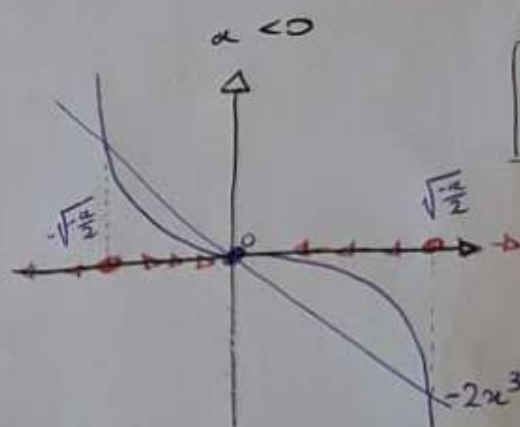
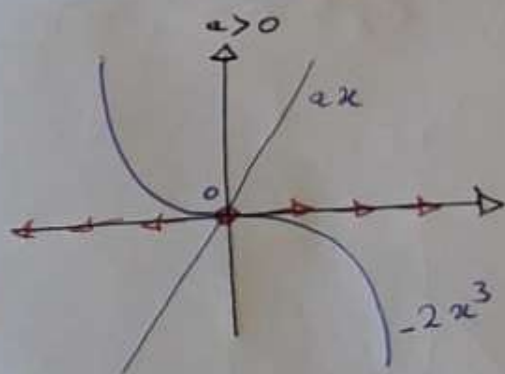
$$x = 0$$

$$a + 2x^2 = 0 \Rightarrow x = \pm \sqrt{\frac{-a}{2}} \quad (\text{only if } a < 0)$$

stability

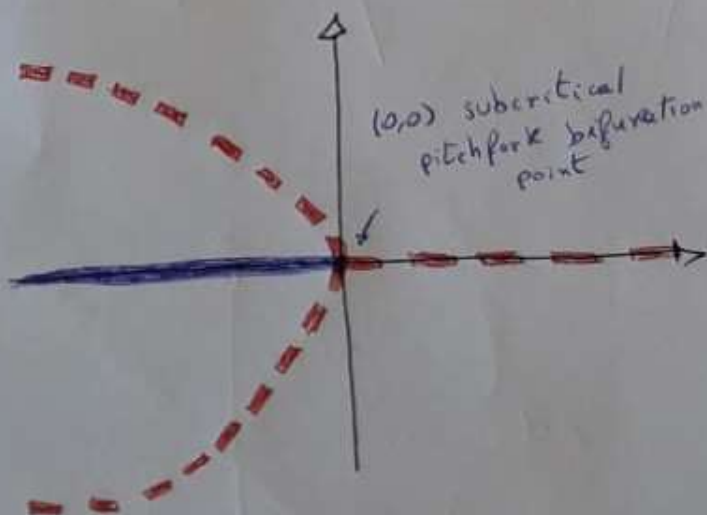
$ax + 2x^3 > 0 \Rightarrow$ trajectory goes right ($ax > -2x^3$)

$ax + 2x^3 < 0 \Rightarrow$ trajectory goes left ($ax < -2x^3$)



• : unstable
• : stable

bifurcation diagram



--- : unstable
— : stable

3. $\frac{dx}{dt} = a + x^2$

equilibrium points

$$a + x^2 = 0$$

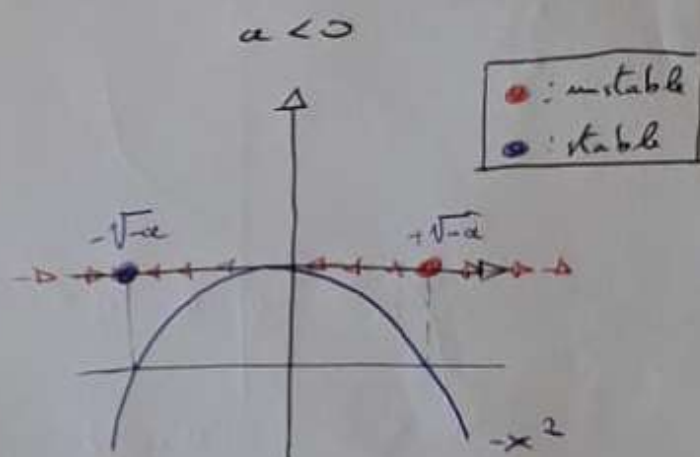
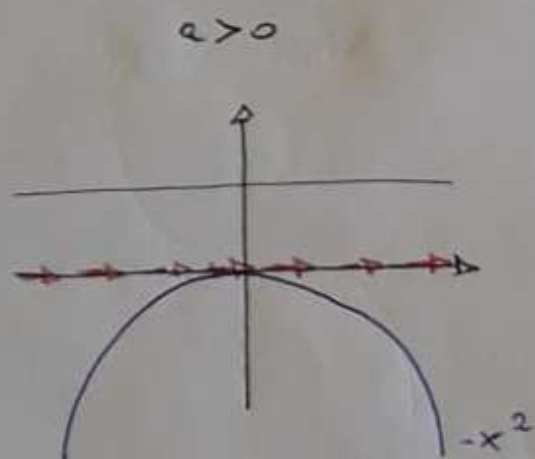
$$x^2 = -a$$

$$x = \pm\sqrt{-a} \quad (\text{only if } a < 0)$$

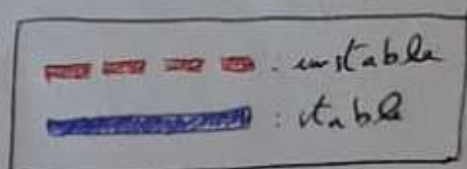
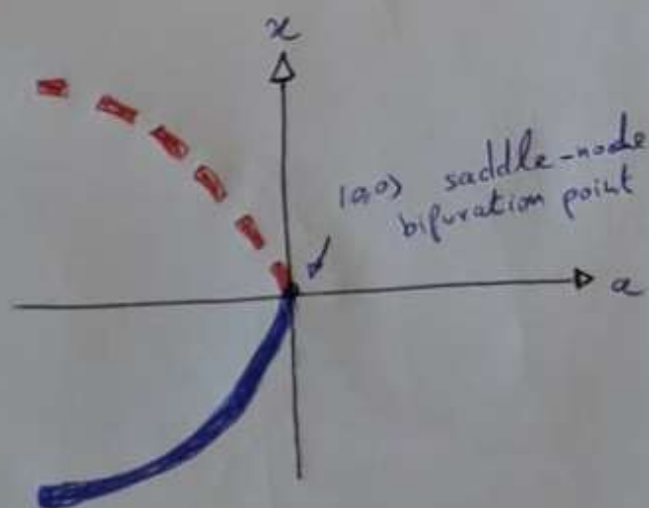
stability

$a + x^2 > 0 \Rightarrow$ trajectory goes right ($a > -x^2$)

$a + x^2 < 0 \Rightarrow$ trajectory goes left ($a < -x^2$)



bifurcation diagram



4.

$$\frac{dx}{dt} = ax - 4x^3$$

equilibrium points

$$ax - 4x^3 = 0$$

$$(a - 4x^2)x = 0$$

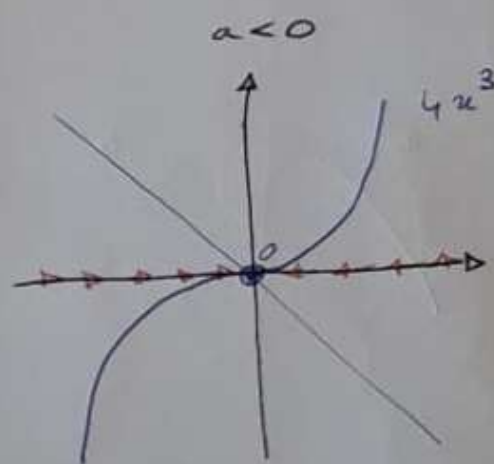
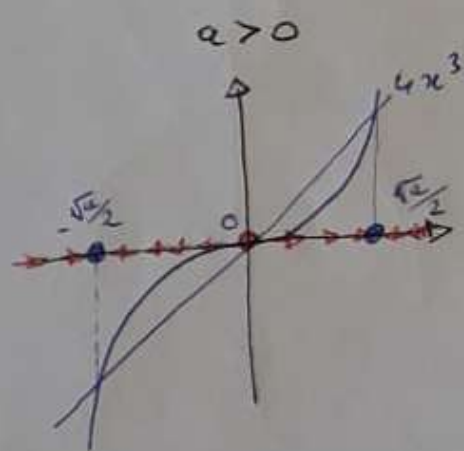
$$x = 0$$

$$a - 4x^2 = 0 \Rightarrow 4x^2 = a \Rightarrow x = \pm \frac{\sqrt{a}}{2} \quad (\text{only if } a > 0)$$

stability

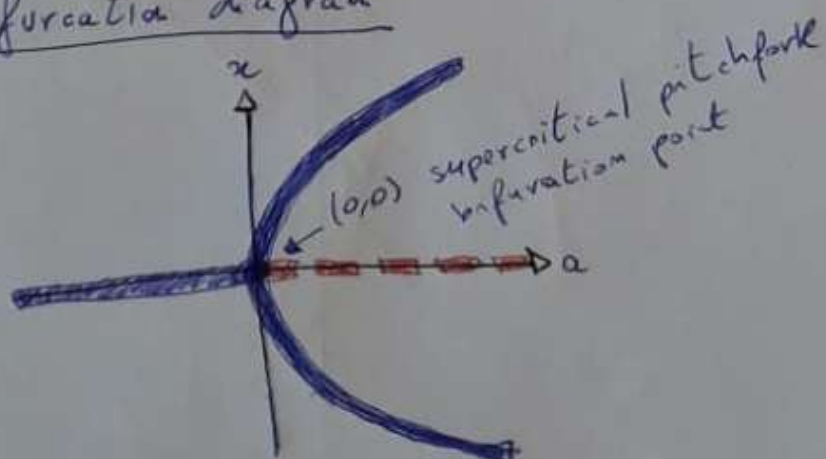
$ax - 4x^3 > 0 \Rightarrow$ trajectory goes right ($ax > 4x^3$)

$ax - 4x^3 < 0 \Rightarrow$ trajectory goes left ($ax < 4x^3$)



• unstable
• stable

bifurcation diagram



--- unstable
— stable

5. $\frac{dx}{dt} = 1 - ax^2$

equilibrium points

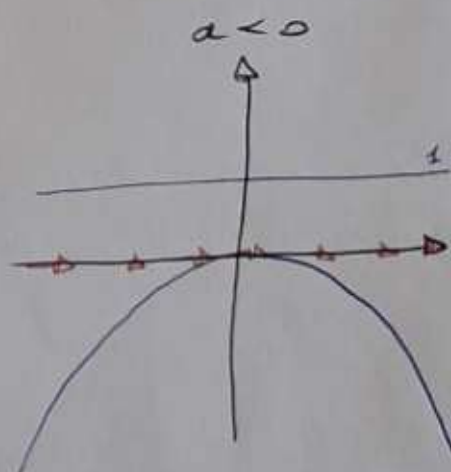
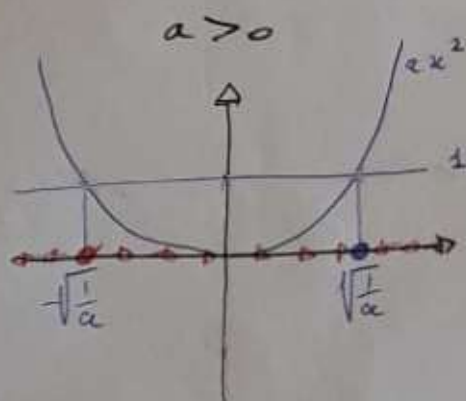
$$1 - ax^2 = 0$$

$$x = \pm \sqrt{\frac{1}{a}} \quad (\text{only if } a > 0)$$

stability

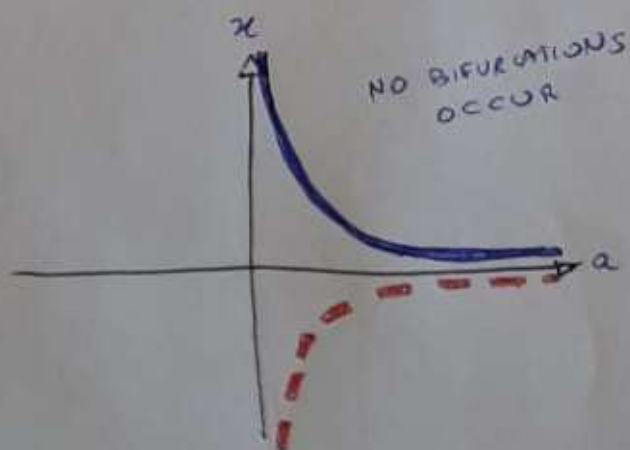
$1 - ax^2 > 0 \Rightarrow$ trajectory goes right ($1 > ax^2$)

$1 - ax^2 < 0 \Rightarrow$ trajectory goes left ($1 < ax^2$)



● : unstable
● : stable

bifurcation diagram



--- : unstable
— : stable